Proof of Lemma 1

Proof. Fix the local states and sampling seeds, and suppress the client and round indices. For $w_0 = w^t$, local training follows $w_{e+1} = w_e - \eta[\nabla\phi_e(w_e+n) + \nabla r(w_e - w^t)]$, where $\phi_e = \ell_{sup} + \mu\ell_{con}$ and $r(v) = \lambda \log(1 + \exp((\|v\|^2 - C^2)/\tau_2))$. Assume $\|\nabla^2\phi_e\|_2 \le L_\phi$, $\lambda \ge 0$, and $\tau_2 > 0$. Let $A_e = \nabla^2\phi_e(w_e + n)$, $H_e = \nabla^2 r(w_e - w^t)$, and $J_e = D_n w_e$. Differentiation gives

$$J_{e+1} = [I - \eta(A_e + H_e)]J_e - \eta A_e, \quad J_0 = 0. \tag{1}$$

For $u = (\|v\|^2 - C^2)/\tau_2$ and $p = (1 + e^{-u})^{-1}$,

$$\nabla^2 r(v) = \frac{2\lambda p}{\tau_2} I + \frac{4\lambda p(1-p)}{\tau_2^2} vv^T. \tag{2}$$

Using $p(1-p) \le 1/4$ and $up(1-p) \le 1/\mathrm{e}$ for $u \ge 0$ gives $\|H_e\|_2 \le L_r$, where $L_r = 2\lambda/\tau_2 + \lambda C^2/\tau_2^2 + 4\lambda/(\mathrm{e}\tau_2)$. Consequently,

$$\|J_{e+1}\|_2 \le [1 + \eta(L_\phi + L_r)]\|J_e\|_2 + \eta L_\phi.$$

Iteration yields $\|J_E\|_2 \le \eta L_\phi \sum_{k=0}^{E-1}[1 + \eta(L_\phi + L_r)]^k$. Since projection onto $B_C = \{v : \|v\| \le C\}$ is nonexpansive, the clipped response $g_i^t(n) = \Pi_{B_C}(w_E(n) - w^t)$ satisfies

$$\mathrm{Lip}(g_i^t) \le \eta L_\phi \sum_{k=0}^{E-1}[1 + \eta(L_\phi + L_r)]^k. \tag{3}$$

□

Proof of Lemma 2

Proof. Let $Z \sim \mathcal{N}(0, \sigma^2 C^2 I_d/m)$ and $F(z) = z + g(z)$, with $\|g(z)\| \le C$ and $\mathrm{Lip}(g) \le \kappa < 1$. For every y, the contraction $x \mapsto y - g(x)$ has a unique fixed point. Thus F is a bi-Lipschitz bijection with Jacobian singular values in $[1-\kappa, 1+\kappa]$ almost everywhere. For $T = F^{-1}$, $\|T(z) - z\| = \|g(T(z))\| \le C$. Writing p_F, p_0 for the output and Gaussian densities, change of variables gives

$$\begin{aligned} &\max\left\{\log\frac{p_F(F(z))}{p_0(F(z))}, \log\frac{p_0(z)}{p_F(z)}\right\} \\ &\quad\le -d\log(1-\kappa) + \frac{m\|z\|}{\sigma^2 C} + \frac{m}{2\sigma^2}, \end{aligned} \tag{4}$$

where the reverse loss uses $\log(1+\kappa) \le -\log(1-\kappa)$. Gaussian concentration yields $\mathbb{E}\|Z\| \le \sigma C\sqrt{d/m}$ and, for $c \ge 0$,

$$\log \mathbb{E}e^{c(\|Z\| - \mathbb{E}\|Z\|)} \le \frac{c^2\sigma^2 C^2}{2m}.$$

For $\alpha > 1$, take the $(\alpha-1)$-exponential moment in Eq. (4) with $c = (\alpha-1)m/(\sigma^2 C)$. In either direction, the Rényi divergence is at most

$$-d\log(1-\kappa) + \frac{\sqrt{md}}{\sigma} + \frac{\alpha m}{2\sigma^2}. \tag{5}$$

□

Proof of Lemma 3

Proof. Let I_t be independent Bernoulli participation indicators with mean q. Condition on a data-independent random index and the sampling schedule. Suppose the conditional round-t RDP bound is $I_t\rho_t(\alpha)$ uniformly over public histories, where ρ_t is deterministic. Adaptive composition bounds the conditional Rényi moment by $\exp((\alpha-1)\sum_t I_t\rho_t(\alpha))$. Averaging over the common index law and the independent indicators, then marginalizing the conditioned variables, bounds the transcript moment by

$$\prod_{t=0}^{T-1}[1 - q + qe^{(\alpha-1)\rho_t(\alpha)}].$$

Taking logarithms and dividing by $\alpha - 1$ gives

$$\rho_T(\alpha) = \sum_{t=0}^{T-1} \frac{\log[1 - q + qe^{(\alpha-1)\rho_t(\alpha)}]}{\alpha - 1}. \tag{6}$$

For neighboring transcript laws P, Q, let $L = \log(dP/dQ)$. The RDP bound and Markov's inequality give

$$P\left[L > \rho_T(\alpha) + \frac{\log(1/\delta)}{\alpha-1}\right] \le \delta.$$

Splitting any event according to this privacy-loss threshold therefore gives (ϵ, δ)-DP, with

$$\epsilon = \inf_{\alpha>1}\left\{\rho_T(\alpha) + \frac{\log(1/\delta)}{\alpha-1}\right\}. \tag{7}$$

□

Proof of Lemma 4

Proof. Fix the selected clients, retained states, and local seeds. For independent $n_i \sim \mathcal{N}(0, \sigma^2 C^2 I_d/m)$, set $\xi_i(n_i) = g_i(n_i) - g_i(0) + n_i$ and $K_i = I + D_n g_i$. Write

$$b = \mathbb{E}_n\left[\frac{1}{m}\sum_i \xi_i\right], \qquad \chi = \frac{1}{m}\sum_i \mathbb{E}_n\|K_i\|_F^2.$$

For the standard Gaussian Ornstein–Uhlenbeck semigroup P_u,

$$\begin{aligned} \mathrm{Var}(v) &= 2\int_0^\infty \mathbb{E}\|\nabla P_u v\|^2 du, \\ \nabla P_u v &= e^{-u} P_u \nabla v. \end{aligned}$$

Jensen's inequality and Gaussian scaling imply

$$\mathrm{Var}(v(n_i)) \le \frac{\sigma^2 C^2}{m}\mathbb{E}_n\|\nabla v(n_i)\|^2.$$

Applying this coordinatewise to ξ_i and summing the independent variances yields

$$\begin{aligned} \mathbb{E}_n\left\|\frac{1}{m}\sum_i \xi_i - b\right\|^2 &\le \frac{\sigma^2 C^2}{m^3}\sum_i \mathbb{E}_n\|K_i\|_F^2 \\ &= \frac{\sigma^2 C^2 \chi}{m^2}. \end{aligned} \tag{8}$$

For $P = P^T$, $0 \le P \le I$, and $\|K\|_2 \le 1$, expanding the square gives

$$\begin{aligned} d - \|I - P + PK\|_F^2 = {} & \mathrm{tr}[P^2(I - KK^T)] \\ & + 2\,\mathrm{tr}[P(I-P)(I - \mathrm{Sym}\,K)] \ge 0, \end{aligned} \tag{9}$$

where $\mathrm{Sym}\,K = (K + K^T)/2$. Both traces are nonnegative because P^2, $P(I-P)$, $I - KK^T$, and $I - \mathrm{Sym}\,K$ are positive semidefinite. □